Eur J Plant Pathol (2019) 155:1195–1209
https://doi.org/10.1007/s10658-019-01848-2

Characterization of tomato mosaic virus and search for its resistance in Solanum species

Najeeb Ullah · Khalid Pervaiz Akhtar · Muhammad Yussouf Saleem · Mudasser Habib

Accepted: 13 September 2019 / Published online: 29 October 2019
© Koninklijke Nederlandse Planteziektenkundige Vereniging 2019

Abstract Tomato mosaic virus (ToMV) is a consistent problem threatening tomato production worldwide, including Pakistan. In this paper, we report characterization of a ToMV strain from naturally infected tomato in Pakistan and response of 128 diverse tomato genotypes to identify potential resistance sources in *Solanum* and its related wild species that can be incorporated into breeding programs. Upon ToMV inoculation, local necrotic lesion developed within 1 wk on plants of *Nicotiana tabacum* var. xanthi, *N. tabacum* var. glutinosa, *Chenopodium album and Datura alba* except *N. tabacum* var. K-399 which showed systemic mosaic symptoms. Coat protein (CP) sequence of ToMV strain (GenBank accession No. MG975645) in present study showed a high level of similarity with the sequence of different ToMV strains submitted in GenBank NCBI. Of 128 tomato genotypes inoculated with ToMV-Pak in controlled conditions, 13 were highly resistant, three resistant, 24 tolerant, 62 susceptible and 26 highly susceptible against ToMV. Among highly resistant germplasm, one genotype (LO6170 with unknown ToMV-resistant gene) and 11 genotypes with ToMV-resistant gene (*Tm-2²*) belonged to cultivated *Solanum lycopersicum* whereas one genotype (LA0111) belonged to the wild species, *S. arcanum*. Presence of *Tm-2²* in the 11 highly resistant genotypes was confirmed using CAPS marker in PCR. Of resistant genetic resources, LB2 belonged to *S. lycopersicum,* LO6122 to *S. arcanum* and LO4166 belonged to *S. pimpinellifolium.* Presence or absence of ToMV in all the tested genotypes plants was confirmed using ELISA and its absence in symptomless highly resistant genotypes with negative ELISA results were further confirmed using RT-PCR. Resistant genotypes of *S. lycopersicum* identified in the present study can be easily used for heterosis and recombination breeding aiming to develop high yielding and ToMV tolerant cultivars. However, wild genotypes resistant to ToMV can be used in a wide hybridization breeding programme with cultivated *S. lycopersicum* keeping in view the genetic barriers and appropriate pattern of crossing.

Keywords Tomato mosaic virus · *Tm-2²* · Hypersensitive response · Wild tomato · RT-PCR

N. Ullah (✉) · K. P. Akhtar · M. Y. Saleem · M. Habib
Nuclear Institute for Agriculture and Biology, Faisalabad, Pakistan
e-mail: najeeb241@hotmail.com

Introduction

Tomato (*Solanum lycopersicum* L.*)* is an important commercial crop of Pakistan, cultivated for vegetable and commercial purposes (Hameed et al. 2017). However, this crop has not yet achieved its potential yield in the country due to a number of biotic and abiotic constraints (Akhtar et al. 2016; Tariq et al. 2018). Among the biotic agents, Tomato mosaic virus (ToMV) particularly reduces quality and production of tomato inflicting economic losses which forces growers to cultivate other crops. Tomato can be infected by about 136 viral species (Hanssen et al. 2010). Among the tomato viruses, *Tomato mosaic virus* (Ullah et al. 2017),

Cucumber mosaic virus (Akhtar et al. 2010) and *Tomato leaf curl New Delhi virus* (Aslam et al. 2017) are the most economically important ones in Pakistan. Among these viruses, *Tomato mosaic virus* (ToMV) known as one of the most prevalent and devastating viruses in tomato worldwide is a rigid rod particle (ssRNA genome) belonging to the genus *Tobamovirus,* family *Virgaviridae* (Nono-Womdim et al. 1996; Khan 1997; King et al. 2011). Yield losses due to ToMV vary depending on the tomato variety infected, the time of infection, virus isolate involved, and the prevailing environmental conditions. Up to 25% yield is reduced due to ToMV infection (Di Candilo et al. 1992), however, early infection can cause greater yield losses (Otieno 1985). ToMV is commonly transmitted through mechanical inoculation, grafting, and most importantly through contaminated seeds (Buchen-Osmond 2006). The infected tomato plants shows yellowing, mosaic, mottling, malformation, necrosis, fern-leaf and shoestring symptoms (Chitra et al. 1999; Hoon and Jin 2002). Being contagious in nature, ToMV is considered as a potential threat particularly for greenhouse tomato production. Therefore, most of the management strategies such as sanitation, rouging of symptomatic plants, and cross protection do not provide efficient control. Tomato breeding strategies have primarily focused on the introgression of resistance alleles from related wild germplasm. The use of genetic resistance has proved successful to combat the problem of ToMV in tomato. Generally, resistant varieties are an effective, durable, economical and environment friendly approach to manage plant diseases (Strange and Scott 2005). In this respect, three dominant ToMV-resistant genes (*Tm-1*, *Tm-2* and *Tm-2*^{2}) were identified in wild tomato species and have been transferred by extensive breeding to develop resistance against ToMV in cultivated tomato (Pelham 1966; Hall 1980; Pfitzner 2006). The *Tm*-1 is an incompletely dominant gene identified in *S. habrochaites* (formerly *Lycopersicon hirsutum*), whereas *Tm*-2 and *Tm*-2^2 identified in *S. peruvianum* (formerly *L. peruvianum)* are considered allelic and confer dominant complete resistance (Pfitzner 2006).

Among different *Tm* genes, *Tm*-2^2 confers resistance to ToMV strains 0, 1 and 2 (Shi et al. 2011). Protein encoded by *Tm*-2 and *Tm*-2^2 recognizes and target movement protein (MP) of ToMV (Lanfermeijer et al., 2003; Weber et al., 2004). For resistant genes confirmation without phenotypic evaluations, DNA-based markers are proved to be a valuable tool. In tomato, several flanking markers are reported linked to different Tm genes and are useful for the selection of resistance in tomato breeding programs (Dax et al. 1998; Ohmori et al. 1996). Recently, a robust and useful *Tm*-2^2 sequence-based molecular CAPS maker are reported by Panthee et al. (2013) for the confirmation of *Tm*-2^2 in tomato genotypes.

Host plant resistance breakdown facilitates the development of plant virus epidemics, making urgent the identification of novel resistance sources in the wild gene pool. In the case of ToMV, concerns on the effectiveness of the resistant genes (*Tm*-1, *Tm*-2 and *Tm*-2^2) have been reported due to appearance of a number of *Tm-1, Tm-2 and Tm-2*^{2} resistance breaking ToMV strains (Pfitzner 2006; Tanwir 2012). Therefore, it is inevitable to test the resistance level of available tomato germplasm against the most prevalent strain of ToMV in the region and to continuously search for effective novel sources of resistance. Moreover, a breeding program can be designed for the best utilization of such identified resources. No information is available on ToMV strains in Pakistan, nor on the response of tomato harboring the most durable resistant gene *Tm-2*^{2}. Therefore, we conducted a comprehensive study to first characterize ToMV strains in Pakistan based on indicator host assay and coat protein sequence (CP) and then screen diverse tomato genotypes including those possessing *Tm-2*^{2} gene.

Materials and methods

Characterization of ToMV strain

Strains of ToMV were recovered from diseased tomato plants showed typical mosaic symptoms in tomato fields at Nuclear Institute for Agriculture and Biology (NIAB), Faisalabad, Pakistan. Leaf tissues from each plant of test genotype were first tested by ELISA (Clark and Adams 1977) for the presence of *Cucumber mosaic virus* (CMV; Agdia USA), *Tomato mosaic virus* (ToMV; Agdia USA) and *Tomato leaf curl New Delhi virus* (ToLCNDV; (AS-1109; DSMZ, Germany) infection as per manufacturer instruction.

ToMV was recovered (from samples which were positive only against ToMV in ELISA), maintained and propagated on plants of susceptible tomato genotype (Nagina) in an insect-proofed glass house. In addition to ELISA, the identification of ToMV was

confirmed by indicator host assay, through reverse transcription polymerase chain reaction (RT-PCR) and sequencing of ToMV CP region.

For indicator host assay, *Nicotiana tabacum* var. xanthi, *N. tabacum* var. glutinosa, *N. tabacum* var. K-399, *Chenopodium album and Datura alba* plants were germinated and maintained in an insect-proofed glasshouse. Six plants of each plant species were mechanically inoculated with ToMV at the 2- to 3-leaf stage. Briefly, ToMV inoculum was prepared by crushing ToMV infected Nagina plant leaves in phosphate buffer (0.01 M, pH 7) at 1:10 (*w/v*) and passed through muslin cloth. Carborundom dusted indicator host plants were then gently inoculated with a cotton swab soaked in ToMV inoculum prepared. Mock inoculation was done with phosphate buffer only. After inoculation, plants were washed with distilled water, observed daily and kept in an insect-proofed glasshouse. To determine ToMV infection, the inoculated plants were observed daily to record symptoms development on the inoculated leaves and the upper systemic leaves. The presence of ToMV on the systemic leaves were also tested using Direct Antigen Coating-Enzyme Linked Immunosorbent Assay (DAC-ELISA). For molecular confirmation, total RNA of ToMV-infected Nagina tomato plant was extracted using TRIzol reagent according to the manufacturer's instructions (Invitrogen, USA). Total RNA extracted was reversed transcribed to cDNA using Fermentas Revert aid first strand cDNA synthesis kit (Fermentas, Lithuania) as per manufacture's protocol. The synthesized cDNA was used as template in RT-PCR to amplify ToMV CP in PCR mixture (20 μl) containing 10 μl of 2X topsimple DyeMIX-nTaq (enzynomics, Korea), 1 μl of cDNA, 0.5 μl (5 pmol/ul) each of the forward (5′-GTTC AGGCGGAAGGCCTAAACC-3′) and reverse primer (5′-AACTTTATATTTCAGCACCTATGCAT-3′) and 8 μl nuclease free water. The initial denaturation was performed at 94 °C for 4 min, followed by 34 cycles of denaturation at 94 °C for 30 s, annealing at 54 °C for 45 s and extension at 72 °C for 1 min. PCR reaction was terminated by final extension at 72 °C for 10 min. The RT-PCR products were separated in agarose gel (1.5%), stained with ethidium bromide and observed using gel documentation system (UVIpro). After eluting the DNA band of interest from the gel using SV gel extraction kit (Korea), one amplicon was directly sequenced in both directions (with the same primers used for amplification) by the dideoxy chain termination method, using a 377 DNA sequencer (ABI, USA). The obtained nucleotide sequences were analysed using BioEdit software and was then compared with the available sequences in BLASTn, NCBI GenBank (http://www.ncbi.nih.gov/gorf/gorf. html). The consensus sequence was obtained which were subsequently deposited in NCBI GenBank (accession No. MG975645). Reference sequences of ToMV were retrieved from GenBank and were aligned together with consensus sequences of our isolate (MG975645) using ClustalW program (Edgar 2004). Phylogenetic analysis was conducted using MEGA7 (Kumar et al., 2016) and a tree was constructed using neighbour-joining method with 1000 pseudoreplicates.

After ToMV strain confirmation using different approaches (ELISA, Indicator host assay, RT-PCR and CP sequence), this strain was multiplied on susceptible tomato genotype (Nagina) and further it was used in the tomato screening experiment.

Screening of tomato genotypes

Core collection of tomato germplasm comprising 128 genotypes belonging to different *Solanum* species was evaluated against ToMV in an insect-proofed glasshouse at 24 ± 2 °C temperature, 14/10 h photoperiod and 50–70% relative humidity. Plant material includes 89 *Solanum lycopersicum* genotypes; one mutant (RioMutant-400 a derivate of determinate cv. Riogrande through irradiation breeding at NIAB, Faisalabad); 11 accessions with ToMV resistant *Tm*-2^2 gene; three accessions of *S. galapagense*; one of *S. neorickii*; three of *S. arcanum*; five of *S. pimpinellifolium*; five *S. pennellii* introgression lines [*S. pennellii* (LA0716) in the background of cv. M-82 LA3475)]; three *S. habrochaites* introgression lines [*S. habrochaites* (LA1777) in the background of cv. E-6203 (LA4024)] and seven backcross recombinant inbred lines (*S. lycopersicum* E-6203 × *S. pimpinellifolium* LA1589). The experimental tomato genotypes of the current investigation were collected from Pakistan, The World Vegetable Center, Taiwan, and the Tomato Genetic Resources Centre (TGRC), USA, and genotypes developed/maintained at the Nuclear Institute for Agriculture and Biology (NIAB), Faisalabad, Pakistan. Nursery seedling of each genotype was raised in pots and 1 mo post-germination plants were thinned to four plants per pot. At the age of 6 weeks, test plants were inoculated using mechanical inoculation keeping variety Nagina as the susceptible control. Top fully-expanded leaves pre-dusted lightly

with carborundom were inoculated with ToMV infected leaf sap (triturated in phosphate buffer; 0.01 M, pH 7) and subsequently washed after 10 min with distilled water. Normal horticultural practices were adapted to keep the plants in good condition. The experimental unit was observed daily and data were recorded on the percentage of disease transmission and latent period (average time required for first symptom appearance after inoculation). Final disease severity at ~50 days post inoculation (DPI) using the modified 0 to 4 disease severity index (DSI) described by Akhtar et al. (2010) given in Table 1, where: 0 indicates no ToMV symptoms; and 4 indicates severe ToMV symptoms, typical mosaic or mottling, leaf deformity, shoe-stringing, stunting with no or few unmarketable fruit setting. Leaf samples from each test genotype plants were collected and tested for ToMV presence using DAC-ELISA as described by Ullah et al. (2017). To maintain uniformity for all the test genotype plants, 1 g (as a standard weight) of newly emerged leaf samples were triturated in the presence of extraction buffer (1:10 *w/v*) and 100 µl of the extracted sap was loaded per well including positive and negative control. The microtiter plates were incubated in a humid box overnight at 4 °C and were washed after each step three times with washing buffer. Then 100 µl of diluted ToMV specific antiserum (1:2000 ratio) was used per well and incubated at 37 °C for 3 h. After plates washing, goat anti-rabbit conjugate was diluted (1:2000 ratio) in antibody buffer and 100 µl was loaded per well followed by incubation for 2 h at 37 °C. In the final step, 100 µl of p-nitrophenyl phosphate (PNPP) dissolved in substrate buffer (1 mg/ml) were added to each well and the plates were kept at room temperature in the dark. The ELISA test plates were observed after substrate addition for colour development and OD reading using ELISA microplate reader (Platos R496, AMP Diagnostic, BSE Austria). Samples were rated as ToMV infected when yellow colour developed in ELISA and its optical density (OD) value exceeded twice that of the negative control. Reaction of *Solanum* species highly resistant to ToMV was further verified in the second season to confirm their true resistant expression.

Table 1 Disease scale for rating of tomato genotypes against ToMV

Rating	Symptoms	Severity index	Disease reaction
0	No visible disease symptoms	0	Highly resistant
1	Complete absence of symptoms or mild mosaic or mottling or leaf deformity, virus can be detected in plant tissues	0.01–1.4	Resistant
2	Moderate mosaic or mottling and leaf deformity followed by minor shoe stringing	1.5–2.4	Tolerant
3	Severe mosaic or mottling, and leaf deformity, shoe-stringing, minor to medium stunting with minor flower shedding and minor reduction in fruit setting but marketable fruit setting	2.5–3.4	Susceptible
4	Severe mosaic or mottling, leaf deformity, shoe-stringing, stunting with no or few unmarketable fruit setting.	3.5–4.0	Highly susceptible

Molecular confirmation of $Tm2^2$ in HR genotypes

For confirmation of *Tm*-2^2 in the highly resistant genotypes of AVRDC in the present study was confirmed using cleaved amplified polymorphic sequences (CAPS) markers linked to *Tm*-2^2 gene (Panthee et al. 2013). Genomic DNA was extracted from young leaves of tomato plants using CTAB based DNA extraction method (Doyle, 1987). The extracted DNA concentration was adjusted to 20 ng/ µl for PCR analysis. PCR was conducted in 20 µl volume consisting of 10 µl 2X topsimple DyeMIX-nTaq (enzynomics, Korea), 2 µl DNA, 0.5 µl each of the forward (5′-AATT TGGGCATACTGACATC-3′) and reverse primer (5′-GTTGCACACATTGGTTGTAG-3′) and 7 µl of nuclease-free water. PCR was performed for an initial denaturation at 94 °C for 4 min, followed by 33 cycles (denaturation at 94 °C for 30 s, annealing at 54 °C for 30 s and extension at 72 °C for 1 min) and a final extension at 72 °C for 10 min. The PCR products were first separated on 1.5% agarose gel and DNA bands were visualized with ethidium bromide and observed using a gel documentation system (UVIpro). After confirmation of PCR amplification, the amplified PCR products were then digested with *Hae*III restriction enzyme (Cat. No R0108S, New England Biolabs) at 37 °C for 2 h. The *Hae*III digested PCR product was then

separated on agarose gel (2%) and visualized with ethidium bromide.

Results

Plant samples which proved positive against ToMV only in ELISA were subjected to further analysis and for screening of diverse tomato genotypes. Upon ToMV inoculation on different plant species, differences in symptoms development and lesion size formation was observed (Table 2*). Plants of N. tabacum var.* xanthi (Fig. 1a) and *N. tabacum* var. glutinosa (Fig. 1b) developed local necrotic lesion. In case of *N. tabacum* var. K-399, systemic mosaic symptoms were observed ~14 dpi and ToMV was detected in the systemic leaves in DAC-ELISA (Fig. 1e). Similarly, ToMV inoculated leaves of *C. album (*Fig. 1c*) and D. alba (*Fig. 1d), also reacted by local necrotic lesion 5-6dpi. The presence of ToMV was also confirmed by RT-PCR in the infected Nagina tomato samples and approximately 617 bp product was successfully amplified using ToMV CP-specific primers in RT-PCR (Fig. 2), as predicted from the position of the primers in sequence of ToMV. BLASTn analysis of the sequence obtained showed a high similarity (96–99%) with different strains of ToMV. Phylogenetic analysis (using the neighbour joining method) placed our sequence (accession No. MG975645) with ToMV reported earlier from Iran on tomato, accession No. KC914397 (Fig. 3).

Of 128 genotypes of tomato, differences in ToMV symptoms development were observed after inoculation. Symptoms observed were marked necrotic lesions, leaf mosaic, mottling, yellowing, narrowing, distortion, curling, reduction in leaf size and plant stunted growth. Success of disease transmission was 100% in 89 genotypes (as on all the inoculated plants of these genotypes symptoms developed and ToMV presence was confirmed in ELISA) belonging to *S. lycopersicum* except in genotype LO6170. All the inoculated plants of genotype LO6170 were symptomless and no virus was detected through ELISA and PCR (Table 3). One genotype LB2 was resistant with mild symptoms, delayed latent period (LP; 20 days) and low ELISA absorbance value (0.899) as compared to highly susceptible genotypes. Seventeen (17) genotypes were tolerant with moderate symptoms with LP ranging 15 to18 days and ELISA absorbance value of 1.105 to 1.915. Forty-nine (49) genotypes were susceptible and 22 were highly susceptible with severe symptoms, short LP and high ELISA absorbance value. Plants of the susceptible genotypes were found readily infected and showed mosaic symptoms 10 dpi which worsened later into severe mottling, leaf deformation and stunting 20 dpi. However, no increase or decrease in symptoms was observed in resistant and tolerant genotypes till the end of experiment.

Among the 11 *S. lycopersicum* genotypes harboring ToMV resistant *Tm*-2^2 gene, none was found to be infected with ToMV (Table 4). All the plants of these genotypes were highly resistant and did not produce any kind of symptoms even at the end of the experiment. These genotypes also exhibited similar highly resistant response during the second growing season. Such genotypes were also found to be negative for ToMV through ELISA and, therefore further tested using a more sensitive approach (RT-PCR) to check presence of ToMV. Interestingly, no viral amplification was observed in *Tm-*2^2 harboring genotypes, hence indicated a complete absence of systemic infection. The presence of *Tm*-2^2 was also confirmed in these highly resistant *S. lycopersicum* genotypes (Table 4) using CAPS markers. PCR product of 870 bp was amplified in PCR using CAPS markers both in susceptible and highly resistant genotypes (Fig. 4). However, after digestion of PCR product with *Hae*III restriction enzyme (NEB), a single uncleaved intact band of 870 bp was

Table 2 Differential responses of selected host plants inoculated with ToMV

Family	Botanical name	English name	Local name	Symptoms[a]	Latent period (days)	Lesion size (mm)
Amaranthaceae	*Chenopodium album*	Common goosefoot	Bathu	CLL	~5	~1.3
Solanaceae	*Datura alba*	Thorn apple	Sufaid dhatoora	NLL	~6	~2
	Nicotiana tabacum var. xanthi	Tobacco	Tobacco	NLL	~3–4	~2.5
	N. tabacum var. glutinosa	Tobacco	Tobacco	NLL	~6	~1.5
	N. tabacum var. K-399	Tobacco	Tobacco	M	~14	NA

[a] Symptoms observed on inoculated plants, CLL = chlorotic local lesion, NLL = necrotic local lesion, M = mosaic, NA = not available

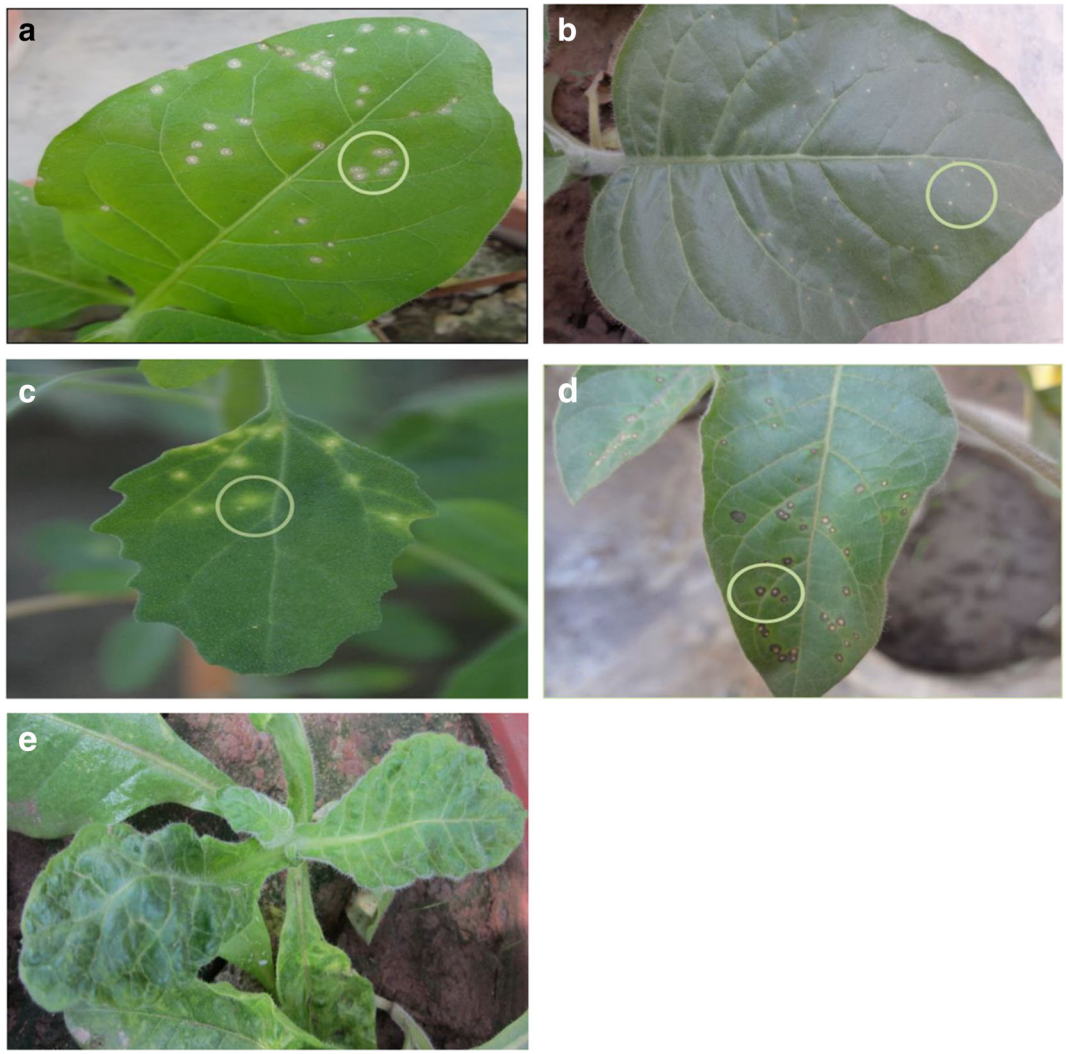


Fig. 1 Differential responses of selected host plants upon ToMV inoculation. Local lesion developed on plants of *Nicotiana tabacum* var. xanthi (**a**), *N. tabacum* var. glutinosa (**b**), *Chenopodium album* (**c**) and *Datura alba* (**d**) while systemic symptoms on *N. tabacum* var. K-399 (**e**)

observed in the susceptible genotypes. Conversely, when the amplified PCR products of all the highly resistant tomato genotypes were digested with *Hae*III restriction enzyme (NEB), it produced two fragments of 270 and 600 bp (Fig. 4) confirming the presence of *Tm*-2^2.

Out of 12 wild accessions of four *Solanum* spp., one accession LA0111 (*S. arcanum*) was highly resistant, one accessions each LO6122 (*S. arcanum*) and LO4166 (*S. pimpinellifolium*) were resistant with mild symptoms, two were tolerant with moderate symptoms, four were susceptible and three were highly susceptible with severe symptoms (Table 5). Entire inoculated plants of highly resistant accession LA0111 were symptomless since no virus was detected through ELISA and RT-PCR. However, disease severity index (DSI) values and LPs for resistant accession LO6122 and LO4166 were 1.0 and 18 days, respectively. ELISA and RT-PCR confirmed the presence of ToMV in them but there was low level of viral accumulation in resistant genotypes as compared to susceptible genotypes when tested through ELISA. Rate of disease transmission was 100% with quite variable SIs, latent periods and virus accumulation in tolerant, susceptible and highly susceptible accessions.

ToMV transmission was 100% in all inoculated plants of introgression and backcross recombinant inbred lines because all lines plants showed different symptoms

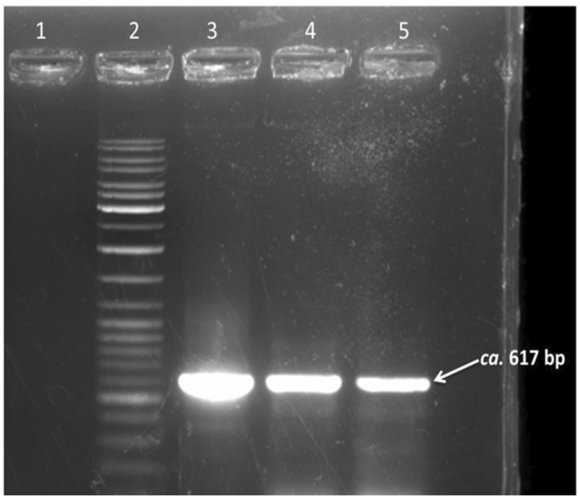


Fig. 2 RT-PCR amplified product of ToMV CP gene from tomato using RT-PCR, Lane 1: healthy tomato plant; Lane 2: 100 bp DNA marker; Lane 3–5: infected tomato plants

(Table 6). None of the inoculated plant was found highly resistant or resistant. Six introgression and backcross recombinant inbred lines were tolerant, eight were susceptible and one was highly susceptible. Appearance of ToMV symptoms started as mild mosaic from 11 to 17 DPI. All the tolerant genotypes developed moderate symptoms while severe symptoms were observed on all the susceptible and highly susceptible genotypes. In these genotypes higher ToMV infection was further confirmed using ELISA (1.341 to 2.432; A_{405nm} 1 h).

Discussion

Accurate identification and early detection of a pathogen is the foremost step in any plant disease management programe to initiate preventive or curative measures (Goyal and Maheshwari 2012; Lacroix et al. 2016). Unfortunately, despite of advances in detection of plant pathogens in Pakistan, no molecular work has been done to detect and characterize ToMV. Therefore, this paper reports for the first time the biological and molecular characterization of ToMV naturally infecting tomato in Pakistan and identification of potential resistance sources in tomato. Local necrotic lesion developed on ToMV inoculated plants of *Nicotiana tabacum* var. xanthi, *N. tabacum* var. glutinosa, *Chenopodium album and Datura alba,* except for *N. tabacum* var. K-399

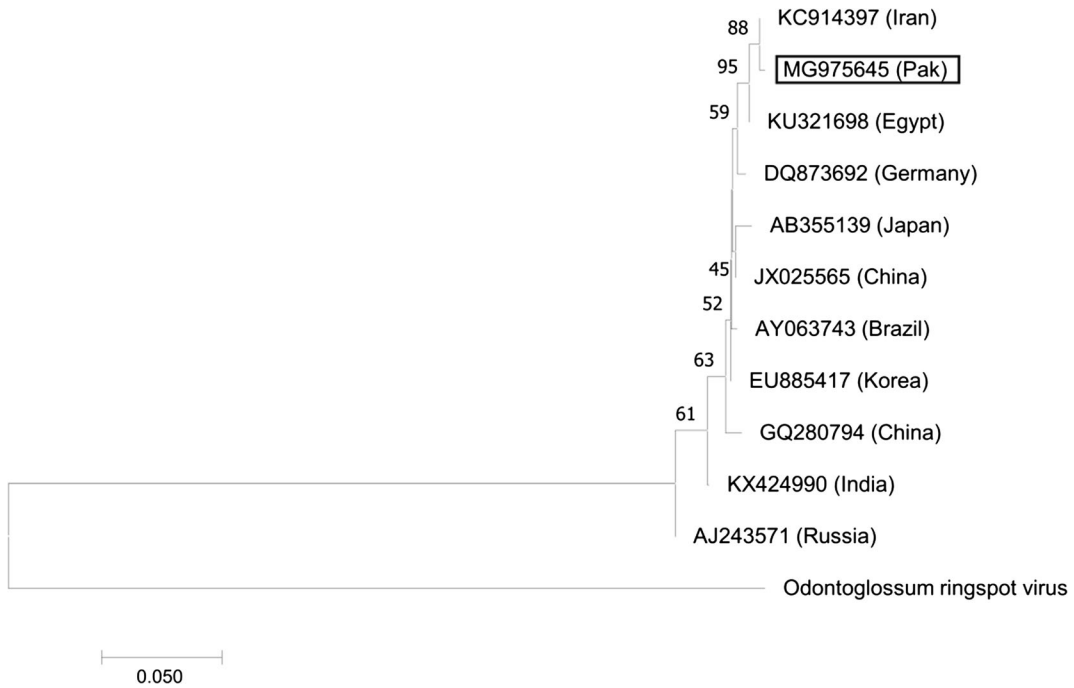


Fig. 3 Phylogenetic tree based on CP sequences showing the relationships between the ToMV-Pak with strains of ToMV reported earlier. The evolutionary history was inferred using the Neighbor-Joining method, with the bootstrap values (1000 replicates) shown next to the branches. Numbers on branches are bootstraps confidence values. Bar, representing evolutionary distance represents 0.01 base substitutions per nucleotide position. The evolutionary distances were computed using the Kimura 2-parameter model using MEGA7. The Odontoglossum ringspot virus (ORSV) was taken as out group

Table 3 Response of different *S. lycopersicum* genotypes against ToMV

Sr. No	Genotype	Type/ Origin or source	LP	SI	DR	ELISA value	Sr. No	Genotype	Type/ Origin or source	LP	SI	DR	ELISA value
1	Alisa Craig	ID/TGRC	11	3.9	HS	2.369	30	LA1226	ID/TGRC	20	3.5	HS	1.751
2	Astra	D/EFUP	18	3.1	S	2.278	31	LA1286	ID/TGRC	15	2.5	S	0.757
3	Beaverlodge	D/TGRC	11	3.8	HS	2.941	32	LA1673	D/TGRC	14	4	HS	1.858
4	B-L-35	D/TGRC	14	2.3	T	1.466	33	LO2875	ID/AVRDC	14	2.8	S	1.954
5	BL1175 (DR3)	D/AVRDC	16	2.8	S	2.469	34	LO6203	D/AVRDC	16	3	S	1.422
6	Bryan Self Topper	D/TGRC	11	3.2	S	3.112	35	LO6170*	D/AVRDC	–	0	HR	0.231
7	Cal Ace	D/TGRC	16	2.8	S	2.682	36	John Bear	ID/TGRC	15	3.3	S	2.842
8	Canada Acc-1	D/EFUP	14	2.5	S	1.199	37	Marglobe	ID/TGRC	11	3.8	HS	3.196
10	Canada-25	D/EFUP	15	2.6	S	2.252	38	Marmande	ID/Bulgaria	11	3.2	S	2.927
11	Jaklin	D/Bulgaria	14	2	T	1.106	39	Meijielo	ID/China	24	3	S	2.538
12	Pak0011009	D/NARC	14	3	S	1.915	40	Money Maker	ID/AARI	18	3.5	HS	1.943
13	BGR13854	ID/Bulgaria	14	2.3	T	2.342	41	Mission102	D/EFUP	24	2	T	1.692
14	CC Haus	D/ AARI	18	3.9	HS	3.182	42	M-82	D/TGRC	16	2.5	S	2.673
15	C-5	D/TGRC	16	3.6	HS	2.925	43	Naqeeb	D/AARI	14	2	T	1.564
16	Earliana	ID/TGRC	14	2.5	S	2.983	44	Nagina	D/AARI	10	3.3	S	2.924
17	Edkawi	D/AVRDC	16	3.5	HS	3.211	45	New Yorker	D/TGRC	12	3.3	S	1.618
18	EFU-06231	SD/AVRDC	16	3.9	HS	2.815	46	Op Boy	D/Holland	16	1.7	T	1.677
19	Fireball	D/TGRC	18	1.5	T	1.323	47	Pakit	ID/ AARI	18	3	S	1.932
20	Floradade	D/TGRC	18	2.8	S	2.603	48	NCEBR-5	D/TGRC	11	2.8	S	1.833
21	Galia	D/GWP	24	2	T	1.229	49	PAK0010575	ID/N. Korea	18	3	S	2.383
22	Gold Nugget	D/TGRC	18	4	HS	3.261	50	PAK0010576	D/N. Korea	24	3	S	1.006
23	Gulf State Market	ID/TGRC	11	3.8	HS	1.614	51	PAK0010983	D/NARC	11	2.5	S	2.433
24	Homestead	ID/TGRC	18	2.7	S	2.382	52	Pak0010994	?/NARC	18	3.4	S	3.233
25	H-24	D/AARI	30	1.5	T	1.392	53	PAK0011000	ID/PARC	11	2.5	S	1.892
26	Islamabad 4–2	D/EFUP	23	1.5	T	1.728	54	Pak0011023	??/NARC	13	2	T	1.105
27	LB2	D/AVRDC	18	1	R	0.899	55	Peto-95-43	D/TGRC	14	2.3	T	1.632
28	LB3	D/AVRDC	20	3	S	2.635	56	Peto-86	D/China	15	3.8	HS	1.769
29	LB4	D/AVRDC	15	2.5	S	2.391	57	Peto	D/Italy	11	2.8	S	2.326
58	Peelo	D/AARI	18	3	S	2.643	75	Tom Round	ID/NARC	11	3.8	HS	2.854
59	Purple Smudge	ID/TGRC	11	3	S	2.295	76	Tibrido	D/ AARI	18	3.5	HS	3.215
60	Red River	ID/TGRC	11	3.6	HS	2.762	77	Titano	D/ AARI	11	3.4	S	3.121
61	Picdenato	D/ AARI	17	3.4	S	1.534	78	Trip-L	ID/USA	18	3.4	S	2.134

Table 3 (continued)

Sr. No	Genotype	Type/ Origin or source	LP	SI	DR	ELISA value	Sr. No	Genotype	Type/ Origin or source	LP	SI	DR	ELISA value
62	Riogrande	D/AARI	14	3.3	S	1.833	79	UC- 82	D/TGRC	18	2.5	S	1.864
63	Riofuego	D/ TGRC	13	2	T	1.856	80	UC- 134	D/TGRC	16	3.4	S	2.354
64	RioMutant-400	ID/NIAB	16	2.7	S	2.738	81	UC-N28	D/TGRC	24	3	S	1.342
65	Roma	D/AARI	14	4	HS	2.916	82	Vender	ID/TGRC	11	3	S	2.463
66	Rockingham	D/NARC	18	2.4	T	1.342	83	Verna Orange	ID/TGRC	11	3	S	3.141
67	Roza	D/TGRC	11	3.3	S	2.935	84	VF-36	D/TGRC	16	3.2	S	2.461
68	Rutgers	ID/TGRC	18	3.5	HS	3.925	85	V-Yaqi	D/ EFUP	14	2.5	S	2.437
69	Saladette	D/TGRC	14	2.8	S	2.383	86	Walter	D/TGRC	18	3	S	2.431
70	Samrudi	D/ EFUP	18	2.5	S	2.211	87	White Beauty	ID/TGRC	11	3.2	S	1.715
71	San Marzana	ID/TGRC	17	4	HS	2.515	88	West Virginia-63	ID/ TGRC	22	1.8	T	1.264
72	Stirling Castle	ID/TGRC	11	3.5	HS	3.323	89	88,572	D/AARI	14	2	T	1.597
73	Stokedale	D/TGRC	16	3.8	HS	3.175	90	09070	?/?	14	2	T	1.915
74	T-2	D/AARI	18	2.6	S	2.321	91	19,291	ID/EFUP	16	4	HS	3.541
	-ive control					0.221							
	+ive control					1.485							

Abbreviations: LPlatent period, SIseverity index, DRdisease response, Ddeterminate, IDindeterminate, SDsemi-determinate, HRHighly resistant, RResistant, TTolerant, SSusceptible, HSHighly susceptible, NIABNuclear Institute for Agriculture and Biology, Pakistan, AARIAyub Agricultural Research Institute, Faisalabad, Pakistan, NARCNational Agricultural Research Council, Pakistan, PARCPakistan Agricultural Research Council, TGRCTomato Genetic Resources Centre, United States of America, AVRDCAsian Vegetable Research and Development Centre, Taiwan, EFUPEstablishment of facilitation unit for participatory vegetable seed and nursery production programme, Pakistan, GWPGujranwala Pakistan, ?not-known, ELISA absorbance values (A_{405nm}) after 1 h, *LO6170 was also negative through PCR

Table 4 Response of *S. lycopersicum* genotypes harboring *Tm-2*^{2} resistant gene against ToMV

Genotype	Internal code	Growth type	LP	DSI	DR	ELISA value	PCR Results	Reference
AVTO1002	CLN3125E	D/AVRDC	–	0	HR	0.331	–	AVRDC
AVTO1003	CLN3125L	D/AVRDC	–	0	HR	0.296	–	AVRDC
AVTO1004	CLN3125O	SD /AVRDC	–	0	HR	0.331	–	AVRDC
AVTO1009	CLN3078G	D/AVRDC	–	0	HR	0.312	–	AVRDC
AVTO1219	CLN3241H-27	SD/AVRDC	–	0	HR	0.301	–	AVRDC
AVTO1311	CLN3241R	SD/AVRDC	–	0	HR	0.209	–	AVRDC
AVTO1315	CLN3241Q	SD/AVRDC	–	0	HR	0.244	–	AVRDC
AVTO9803	CLN1621F	D/AVRDC	–	0	HR	0.317	–	AVRDC
AVTO9809	CLN2037E	?/AVRDC	–	0	HR	0.309	–	AVRDC
AVTO1005	CLN3125P	D/AVRDC	–	0	HR	0.325	–	AVRDC
CL5915–206-D4–2–2-0	–	ID/AVRDC	–	0	HR	0.321	–	Seid et al. 2017

PCR results: - PCR results: - = negative, + = positive; Remaining foot note is same as for Table 2

which showed systemic mosaic symptoms. ToMV is reported to induce local lesion on the leaves of *Nicotiana spp.* carrying N gene (for necrotic lesion response) resulting in hypersensitive response (HR) (Aramburu and Galipienso 2005; Kumar et al. 2011). Generally, HR response due to N gene results into local lesions production and their size reflects the movement of virus in plants, its replication and defense responses of plants (Yu et al. 2004). However, systemic symptoms develop on ToMV infected tobacco plants lacking N gene (Yu et al. 2004). BLASTn analysis of the ToMV sequence in the present study revealed minor differences and shared 96–99% identity with other strains of ToMV. This shows that strain used in our study belongs to ToMV

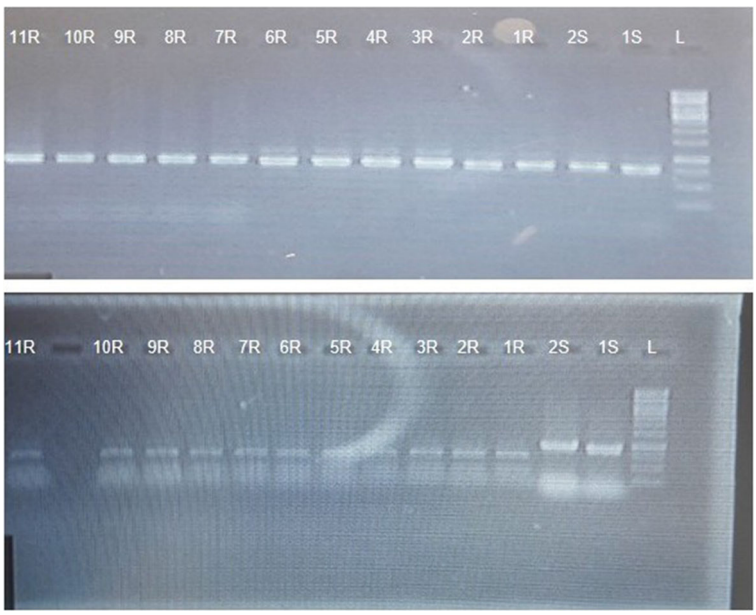


Fig. 4 Screening of 11 highly resistant and two susceptible tomato genotype plants with CAPS molecular marker linked to *Tm*-2^2 gene. *Hae*III restriction enzyme digested PCR product amplified with CAPS primers produce two bands (600 and 270 bp) in highly resistant genotypes, whereas a single uncleaved band (870 bp) in the susceptible genotypes plants. In the upper gel is PCR amplified product with CAPS primers while in the lower gel is the restriction with *Hae*III. L = 1 kb DNA Ladder (GenRuler); 1S = Roma; 2S = Money Maker; 1R = AVTO1002; 2R = AVTO1003; 3R = AVTO1004; 4R = AVTO1009; 5R = AVTO1219; 6R = AVTO1311; 7R = AVTO1315; 8R = AVTO9803; 9R = AVTO9809; 10R = AVTO1005; 11R = CL5915–206-D4–2–2-0

Table 5 Response of different genotypes of *Solanum Spp.* against ToMV

Species/ Entry	Internal code	Type/ Origin or source	LP	SI	DR	ELISA value	PCR Results
Solanum galapagense ***(S. chmielewskii)***							
TGRC-9	LA1261		20	3	S	2.135	NT
TGRC 10	LA0317	D/TGRC	20	3	S	1.355	NT
TGRC 24	LA1306	ID/TGRC	14	3.6	HS	2.173	NT
S. neorickii (S. parviflorum)							
TGRC 14	LA2727	D/TGRC	15	3	S	1.405	NT
S. arcanum (S. peruvianum)							
TGRC 21	LA0111	D/TGRC	–	0	HR	0.250	–
AVRDC 21	LO6221	D/AVRDC	18	1	R	0.523	NT
AVRDC 22	LO6122	ID/AVRDC	19	1.5	T	1.862	NT
S. pimpinellifolium							
TGRC 8	LA0722	D/TGRC	15	2.5	S	1.213	NT
TGRC 7	LA2184	D/TGRC	20	3.7	HS	1.573	NT
AVRDC 25	LO2707	SD/AVRDC	20	4	HS	2.038	NT
AVRDC 24	LO3715	SD/AVRDC	13	2	T	1.179	NT
AVRDC 26	LO4166	ID/AVRDC	18	1	R	0.744	+

PCR results: - = negative, + = positive; NT = Not tested; Remaining foot note is same as for Table 2

as for Tobamovirus species differentiation less than 90% sequence identity at nucleotide level is considered as new specie (Adams et al. 2012). Based on its sequence analysis, the obtained strain is not an emergent strain with ability to break tomato resistance harboring *Tm-2*^{2} gene. Phylogenetic analysis placed ToMV-Pak

Table 6 Response of different *Solanum* introgression lines and backcross recombinant inbred lines against ToMV

Species/ Entry	Type/ Origin or source	LP	SI	DR	ELISA value
S. pennellii introgression lines *(S. pennellii* [LA0716] in the background of cv. M-82 [LA3475])					
LA4054	?/TGRC	17	2.5	S	1.852
LA4064	?/TGRC	17	2.0	T	1.631
LA4071	?/TGRC	19	1.7	T	1.502
LA4099	?/TGRC	17	2.9	S	1.837
LA4146	?/TGRC	20	1.9	T	1.527
S. habrochaites introgression lines (*S. habrochaites* LA1777 in the background of cv. E-6203 [LA4024])					
LA3925	?/TGRC	14	2.8	S	2.543
LA3952	?/TGRC	14	2.0	T	1.832
LA39 60	?/TGRC	17	2.0	T	1.345
Backcross Recombinant Inbred Lines (*S. lycopersicum* E-6203 × *S. pimpinellifolium* LA1589)					
LA4139	?/TGRC	17	2.5	S	1.930
LA4145	?/TGRC	17	3.0	S	1.341
LA4147	?/TGRC	14	2.5	S	1.660
LA4151	?/TGRC	17	3.0	S	1.429
LA4153	?/TGRC	15	3.5	HS	2.432
LA4157	?/TGRC	17	2.0	T	1.418
LA4160	?/TGRC	11	3.2	S	1.632

PCR results: - = negative, + = positive; Remaining foot note is same as for Table 2

(accession No. MG975645) with ToMV reported earlier from Iran on tomato, accession No. KC914397 (Yazdani-Khameneh et al., 2013). Therefore, investigation of Pakistani isolates of ToMV with Iranian virus isolates will provide useful information on the origin and distribution of ToMV within the country.

The search for resistance in tomato against ToMV has always been under discussion, because the virus is aggressive, highly contagious and drastically reduce tomato yield (Tanwir 2012). Due to current uncertainty in climate change, utilization of host resistance to improve disease management has the potential to enhance global food production and security. Furthermore, changes in virus isolate aggressiveness also necessitate testing the resistance level against the strain of the virus prevalent in the region from time to time. For that reason, a comprehensive evaluation of a large number of cultivated and wild genotypes belonging to different Solanum species was performed against Pakistani strain of ToMV using mechanical inoculation. Results showed wide variations in response to ToMV among the examined genotypes as a significant number were found susceptible while few were highly resistant. Of 99 *S. lycopersicum* genotypes, one genotype LO6170 was highly resistant to ToMV as it was symptomless and no virus was detected through ELISA and RT-PCR. One genotype LB2 was resistant with mild symptoms, delayed latent period and low ELISA value as compared to highly susceptible genotypes. Remaining genotypes were tolerant to highly susceptible showing a range of symptoms including yellow spot, necrotic lesion, mosaic, mottling, narrowing, deformation and curling of leaves etc. as reported elsewhere (Chitra et al. 2002; Hoon and Jin 2002; Ullah et al. 2017). In a breeding program, resistance level evaluation of plant genotype correspond to the effect of infection on total yield and yield components. The infection severity and the virus accumulation level can serve as indictors of resistance level (Lapidot et al. 1997; Azizi et al. 2008). In some of the inoculated genotypes mild symptoms and low ToMV titer was observed compared to susceptible control. Based on their phenotypic evaluation and ToMV titer assessment these genotypes were rated as resistant and tolerant. The highly resistant genotypes accumulate low viral titer (Pico et al. 1999). Conversely, the susceptible genotypes show severe symptoms and high viral accumulation (Kuria et al. 2017).

The scarcity of highly resistance sources within the available commercial *S. lycopersicum* genotypes and hybrids enhances the importance of other sources of resistance against ToMV. In this regard, resistance genes found in wild tomato species have been utilized to manage ToMV (Pelham 1966; Hall 1980). So far, in wild *Lycopersicon* species three dominant resistant genes against ToMV namely *Tm-1* (from *L. hirsutum*), *Tm-2* and *Tm-2*^{2} (both from *L. peruvianum*) have been transferred by extensive breeding in to cultivated tomato (Pfitzner 2006). These resistant genes restrict multiplication of ToMV and its systemic movement in tomato plant (Smith and Murakishi 1993). However, soon *Tm-1* gene resistance was compromised by ToMV resistance breaking strains (Motoyoshi and Oshima 1977). After the discovery of *Tm-1* gene, other important resistant gene named as *Tm-2* and *Tm-2*^{2} were identified which provided more effective and durable resistance. *Tm-2*^{2} appears to be an allele of *Tm-2* and both are located on chromosome 9 (Alexander 1971) and are CC -NBS-LRR type of resistance genes. These resistance genes are mostly associated with a hypersensitive reaction (HR) [Weber and Pfitzner 1998; Weber et al. 2004] and HR induced by *Tm-2*^{2} is more extreme than *Tm-2* thus provide more durable resistance (Pelham I972). Recently, the World Vegetable Center developed few fresh market tomato lines resistant to ToMV by the introgression of *Tm-2*^{2} gene. We also evaluated 11 genotypes containing the *Tm-2*^{2} gene against the ToMV strain present in Pakistan. All such genotypes were highly resistant (symptomless) and found negative for ToMV through ELISA. A more sensitive technique (PCR) is recommended for the confirmation of virus presence or absence particularly in highly resistant genotypes, which accumulate low viral titer (Pico et al. 1999). Therefore, inoculated plants of these highly resistant genotypes were further tested using RT-PCR, but ToMV was not detected which confirms that these genotypes are not symptomless carrier.

Several flanking markers are reported for the confirmation of different *Tm* genes and used in tomato breeding programs (Dax et al. 1998; Ohmori et al. 1996). Recently, a CAPS marker was designed from the locus specific sequence for *Tm*-2^2 which can efficiently differentiate the presence or absence of *Tm*-2^2 in tomato genotypes (Panthee et al. 2013). Sequence-based molecular markers are known to be more useful and robust than markers from flanking region (Varshney et al. 2005; Panthee et al. 2013). In the present study, this CAPS marker was used for the confirmation of *Tm*-2^2 in these 11 highly resistant tomato genotypes. The CAPS marker clearly differentiated these 11 highly resistant

genotypes from susceptible ones. The *Tm*-2^2 was only found in these 11 highly resistant genotypes and not in the susceptible genotypes. Furthermore, result of this marker was consistent with phenotypic data. The results showed that the CAPS marker is reliable, robust and specific to *Tm*-2^2 gene and can be used for ToMV resistance confirmation in tomato breeding programs.

Earlier, Zhang et al. (2013) reported that resistance conferred by *Tm*-2^2 includes extreme resistance (ER) and HR determined by expression level of *Tm*-2^2 gene. Furthermore, in *Tm*-2^2 conferring ER, small subunit of RuBisCo was involved (Zhao et al. 2013). The ER response of highly resistant genotypes against ToMV in our present study may be due to resistance responses at the single cell level or due to untraceable pathogen accumulation (Yamaji et al. 2012).

High mutation rates in plant viruses result in the appearance of mutant virus strains from time to time overcoming the plant resistance barriers to increase its survival and spread. Resistance breaking mutant strains of different viruses have been reported intermittently in cotton, tomato, and other crops (Pelham 1972; Mansoor et al. 2003; Strasser and Pfitzner 2007). Recent report of a tobamovirus overcoming *Tm*-2^2 resistance in commercial varieties of tomato is the evidence of such mutation (Luria et al. 2017). Hence, to combat such situation exploration of new sources of resistance is demanded. Most of the resistance genes against viruses are found in wild relatives of the crops, from which they are introduced into cultivated genotypes (Ishibashi et al. 2012). Of 12 wild accessions belonging to four *Solanum* spp., one accession LA0111 from *S. arcanum* was highly resistant while one accession LO6122 from *S. arcanum* and LO4166 from *S. pimpinellifolium* were resistant with mild symptoms followed by confirmation through ELISA and PCR. Therefore, these accessions can possibly be used as a donor parent of resistance against ToMV to widen the genetic base. Although, the cross of these wild sources is difficult with *S. lycopersicum* due to unilateral incompatibility and embryo abortion barriers, these may be overcome through in vitro embryo rescue techniques (Akhtar et al. 2016).

Conclusions

The current study reports the molecular and biological characterization of ToMV strain present in Pakistan for the first time. It advances our understanding about the identification of potential ToMV-resistant sources in tomato. From this study it is evident that ToMV strain in Pakistan is not an emergent strain and *Tm*-2^2 provides efficient resistance. Different genetic resources resistant to ToMV in tomato as an outcome of the study can serve as a potential reservoir. Prospects are positive to breed tomato within a cultivated background, eventually leading to high yielding and ToMV resistant tomato cultivars. However, special breeding techniques or methods would have to be followed if ToMV genes conferring resistance from wild species into cultivated background is intended.

Funding The authors received no specific funding for this work.

Compliance with ethical standards

Conflict of interest The authors declare that they have no competing interests.

Ethical statement This research did not involve human participants and/or animals.

References

Adams, M. J., Heinze, C., Jackson, A. O., Kreuze, J. F., Macfarlane, S. A., & Torrance, L. (2012). Virgaviridae. In A. M. O. King, E. Lefkowitz, M. J. Adams, & E. B. Carstens (Eds.), *Virus taxonomy: Ninth report of the international committee on taxonomy of viruses* (pp. 1139–1162). London: Elsevier/Academic Press.

Akhtar, K. P., Saleem, M. Y., Asghar, M., Ahmad, M., & Sarwar, N. (2010). Resistance of Solanum species to cucumber mosaic virus subgroup IA and its vector Myzus persicae. *Eurpean Journal of Plant Pathology, 128*, 435–450.

Akhtar, K. P., Saleem, M. Y., Iqbal, Q., Asghar, M., Hameed, A., & Sarwar, N. (2016). Evaluation of tomato genotypes for late blight resistance using low tunnel assay. *Journal of Plant Pathology, 98*, 421–428.

Alexander, L. J. (1971). Host pathogen dynamics of Tobacco mosaic virus on tomato. *Phytopathology, 61*(61), 1–617.

Aramburu, J., & Galipienso, L. (2005). First report in Spain of a variant of tomato mosaic virus (ToMV) overcoming the tm-22 resistance gene in tomato (Lycopersicon esculentum). *Plant Pathology, 54*, 566–566.

Aslam, M. Q., Akhtar, K. P., Saleem, M. Y., Sarwar, N., & Yousaf, S. (2017). Association of chili leaf curl betasatellite with tomato leaf curl disease. *Journal of General Plant Pathology, 83*(6), 402–405.

Azizi, A., Mozafari, J., & Shams-bakhsh, M. (2008). Phenotypic and molecular screening of tomato germplasm for resistance to tomato yellow leaf curl virus. *Iranian Journal of Biotechnology, 6*(4), 199–206.

Buchen-Osmond, C. (2006). *ICTVdB—The universal virus database, version 4*. New York: Columbia University.

Chitra, T. R., Prakash, H. S., Albrechtsen, S. E., Shetty, H. S., & Mathur, S. B. (1999). Infection of tomato and bell pepper by ToMV and TMV at different growth stages and establishment of virus in seeds. *Journal of Plant Pathology, 81*, 123–126.

Chitra, T. R., Prakash, H. S., Albrechtsen, S. E., Shetty, H. S., & Mathur, S. B. (2002). Indexing of leaf and seed samples of tomato and bell pepper for tobamoviruses. *Indian Phytopathology, 55*, 84–86.

Clark, M. F., & Adams, A. N. (1977). Characteristics of the microplate method of enzyme-linked immunosorbent assay for the detection of plant viruses. *Journal of General Virology, 34*, 475–483.

Dax, E., Livneh, O., Aliskevicius, E., Edelbaum, O., Kedar, N., Gavish, N., Milo, J., Geffen, F., Blumenthal, A., Rabinowich, H. D., & Sela, I. (1998). A SCAR marker linked to the ToMV resistance gene, Tm2(2), in tomato. *Euphytica., 101*, 73–77.

Di Candilo, M., Faccioli, G., Grassi, G., Faeti, V. (1992). Effect of tomato mosaic virus (ToMV) on yield of machine-harvested processing tomatoes. *Phytopathologia Mediterranea,* 32–36.

Doyle, J. J. (1987). A rapid DNA isolation procedure for small quantities of fresh leaf tissue. *Phytochemical bulletin, 19*, 11–15.

Edgar, R. C. (2004). MUSCLE: Multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Research, 32*, 1792–1797.

Goyal, A., Maheshwari, P. (2012). Accurate diagnosis of a plant disease and its pathogen is the first crucial step for its management. 1:116–136.

Hall, T. J. (1980). Resistance at the tm-2 locus in the tomato to tomato mosaic virus. *Euphytica, 29*, 189–197.

Hameed, A., Saleem, M. Y., Akhtar, K. P., Shoaib, M., Iqbal, Q., & Asghar, M. (2017). Molecular confirmation of intraspecific tomato (Solanum lycopersicum) hybrids and their evaluation against late blight and cucumber mosaic virus. *Molecular Biotechnology, 59*, 234–240.

Hanssen, I. M., Lapidot, M., & Thomma, B. P. (2010). Emerging viral diseases of tomato crops. *Molecular Plan-Microbe Interaction, 23*(5), 539–548.

Hoon, P. K., & Jin, C. B. (2002). Detection of TMV, ToMV and CMV from tomato seeds and plants. *Research in Plant Disease, 8*, 101–106.

Ishibashi, K., Mawatari, N., Miyashita, S., Kishino, H., Meshi, T., & Ishikawa, M. (2012). Coevolution and hierarchical interactions of tomato mosaic virus and the resistance gene tm-1. *PLoS Pathogen, 8*(10), e1002975.

Khan, I. A. (1997). Occurrence, distribution, host range, symptomatology and purification of ToMV on tomato. *Pakistan Journal of Zoology, 29*, 385–389.

King, A.M., Adams, M.J., Lefkowitz, E.J., Carstens, E.B. (2011). Virus taxonomy: IXth report of the international committee on taxonomy of viruses.

Kumar, S., Udaya, Shankar, A. C., Nayaka, S. C., Lund, O. S., & Prakash, H. S. (2011). Detection of tobacco mosaic virus and tomato mosaic virus in pepper and tomato by multiplex RT–PCR. *Letters in Applied Microbiology, 53*(3), 359–363.

Kumar, S., Stecher, G., & Tamura, K. (2016). MEGA7: Molecular evolutionary genetics analysis version 7.0 for bigger datasets. *Molecular Biology and Evolution, 33*(7), 1870–1874.

Kuria, P., Ilyas, M., Ateka, E., Miano, D., Onguso, J., Carrington, J. C., & Taylor, N. J. (2017). Differential response of cassava genotypes to infection by cassava mosaic geminiviruses. *Virus Research, 227*, 69–81.

Lacroix, C., Renner, K., Cole, E., Seabloom, E. W., Borer, E. T., & Malmstrom, C. M. (2016). Methodological guidelines for accurate detection of viruses in wild plant species. *Applied and Environmental Microbiology, 82*, 1966–1975.

Lanfermeijer, F. C., Dijkhuis, J., Sturre, M. J., de Haan, P., & Hille, J. (2003). Cloning and characterization of the durable tomato mosaic virus resistance gene tm-2 2 from Lycopersicon esculentum. *Plant Molecular Biology, 52*(5), 1039–1051.

Lapidot, M., Friedmann, M., Lachman, O., Yehezkel, A., Nahon, S., Cohen, S., & Pilowsky, M. (1997). Comparison of resistance level to tomato yellow leaf curl virus among commercial cultivars and breeding lines. *Plant Disease., 81*, 1425–1428.

Luria, N., Smith, E., Reingold, V., Bekelman, I., Lapidot, M., Levin, I., Elad, N., Tam, Y., Sela, N., Abu-Ras, A., & Ezra, N. (2017). A new Israeli Tobamovirus isolate infects tomato plants harboring tm-22 resistance genes. *PLoS One, 12*(1), e0170429.

Mansoor, S., Amin, I., Iram, S., Hussain, M., Zafar, Y., Malik, K. A., & Briddon, R. W. (2003). Breakdown of resistance in cotton to cotton leaf curl disease in Pakistan. *Plant Pathology, 52*, 784.

Motoyoshi, F., & Oshima, N. (1977). Expression of genetically controlled resistance to tobacco mosaic virus infection in isolated tomato leaf mesophyll protoplasts. *Journal of General Virology, 34*, 499–506.

Nono-Womdim, R., Swai, I. S., Green, S. K., Gebre-Selassie, K., Laterrot, H., Marchoux, G., & Opena, R. T. (1996). Tomato viruses in Tanzania: Identification, distribution and disease incidence. *J. S. Afr. Soc. Hort. Scie., 61*, 41–44.

Ohmori, T., Murata, M., & Motoyoshi, F. (1996). Molecular characterization of RAPD and SCAR markers linked to the tm-1 locus in tomato. *Theoretical and Applied Genetics, 92*, 151–156

Otieno, E. A. (1985). Identification of tomato mosaic strain of Tobacco mosaic virus (TMV) and its effects on yield of tomato (Lycopersicon esculentum) varieties moneymaker and Roma-vf in kenya. *Journal of Agriculture and Environment, 3*, 56–58.

Panthee, D. R., Brown, A. F., Yousef, G. G., Ibrahem, R., & Anderson, C. (2013). Novel molecular marker associated with T m2a gene conferring resistance to tomato mosaic virus in tomato. *Plant Breeding, 132*(4), 413–416.

Pelham, J. (1966). Resistance in tomato to tobacco mosaic virus. *Euphytica, 15*, 258–267.

Pelham, J. (1972). Strain-genotype interaction of tobacco mosaic virus in tomato. *Annals of Applied Biology, 71*, 219–228.

Pfitzner, A.J. (2006). Resistance to tobacco mosaic virus and tomato mosaic virus in tomato. In Natural Resistance Mechanisms of Plants to Viruses. 399–413.

Pico, B., Díez, M. J., & Nuez, F. (1999). Improved diagnostic techniques for tomato yellow leaf curl virus in tomato breeding programs. *Plant diseas, 83*(11), 1006–1012.

Shi, A., Vierling, R., Grazzini, R., Chen, P., Caton, H., & Panthee, D. (2011). Molecular markers for tm-2 alleles of tomato mosaic virus resistance in tomato. *American Journal of Plant Sciences, 2*(2), 180–189.

Smith, S., & Murakishi, H. H. (1993). Restricted virus multiplication and movement of tomato mosaic virus in resistant tomato somaclones. *Plant Science, 89*, 113–122.

Strange, R. N., & Scott, P. R. (2005). Plant disease: A threat to global food security. *Phytopathology, 43*, 83–116.

Strasser, M., & Pfitzner, A. J. P. (2007). The double-resistance-breaking tomato mosaic virus strain ToMV1-2 contains two independent single resistance-breaking domains. *Archives of Virology, 152*, 903–914.

Tanwir, F. (2012). Analysis of the structure of Tomato mosaic virus movement protein based on virus host interactions. PhD Thesis.

Tariq, R.M.S., Akhtar, K.P., Hameed, A., Ullah, N., Saleem, M.Y. Haq, I. (2018). Determination of the role of salicylic acid and Benzothiadiazole on physico-chemical alterations caused by cucumber mosaic virus in tomato. *Eurpean Journal of Plant Pathology,* 1–12.

Ullah, N., Ali, A., Ahmad, M., Fahim, M., Din, N., & Ahmad, F. (2017). Evaluation of tomato germplasm for the source of resistance against tomato mosaic virus (ToMV) and effect of the virus on yield contributing parameters. *Pakistan Journal of Botany, 49*(4), 1585–1592.

Varshney, R. K., Graner, A., & Sorrells, M. E. (2005). Genomics-assisted breeding for crop improvement. *Trends in Plant Science, 10*(12), 621–630.

Weber, H., & Pfitzner, A. J. (1998). Tm-22 resistance in tomato requires recognition of the carboxy terminus of the movement protein of tomato mosaic virus. *Molecular Plant-Microbe Interactions, 11*(6), 498–503.

Weber, H., Ohnesorge, S., Silber, M. V., & Pfitzner, A. J. (2004). The tomato mosaic virus 30 kDa movement protein interacts differentially with the resistance genes tm-2 and tm-2(2). *Archives of Virology, 149*, 1499–1514.

Yamaji, Y., Maejima, K., Komatsu, K., Shiraishi, T., Okano, Y., Himeno, M., Sugawara, K., Neriya, Y., Minato, N., Miura, C., & Hashimoto, M. (2012). Lectin-mediated resistance impairs plant virus infection at the cellular level. *Plant Cell, 24*, 778–793.

Yazdani-Khameneh, S., Hamedi, A., Farahani, A. A., Hashemi, S., Golnaraghi, A. R., & Rakhshandehroo, F. (2013). Tomato mosaic virus on cucumber and potato in Iran. *Journal of Plant Pathology, 95*(3), 659–668.

Yu, C., Hu, D., Dong, J., Cui, X., Wu, J., Yu, J., & Zhou, X. (2004). The symptom difference induced by tobacco mosaic virus and tomato mosaic virus in tobacco plants containing the N gene is determined by movement protein gene. *Science in China Series C: Life Sciences, 47*(6), 503–509.

Zhang, H., Zhao, J., Liu, S., Zhang, D. P., & Liu, L. (2013). Tm-22 confers different resistance responses against tobacco mosaic virus dependent on its expression level. *Molecular Plant, 6*, 971–974.

Zhao, J., Liu, Q., Zhang, H., Jia, Q., & Hong, L. Y. (2013). The Rubisco small subunit is involved in Tobamovirus movement and tm-22-mediated extreme resistance. *Plant Physiology, 161*, 374–383.